NeuroADHD — Quick Start Guide

Zero-to-Execution Manual for Local MRI Preprocessing & Connectivity

RESEARCH PROTOTYPE v2.0

> **RESEARCH SAFETY NOTICE**
> NeuroADHD is an exploratory computational neuroscience tool for resting-state fMRI feature extraction. It is **NOT** a clinical diagnostic device. Processing operates in `STRICTLY_SYNTHETIC_ONLY` mode until full clinical research authorization is configured.

1 System Requirements

Hardware / Software	Requirement	Verified Project Baseline
Operating System	macOS (Darwin ARM64 / M1 / M2 / M3) or Linux	macOS 15 (Apple Silicon M-Series)
Python Version	Python 3.10 or Python 3.11	Python 3.11.15 in isolated `.venv`
System Memory (RAM)	Minimum 4 GB (Recommended: 8 GB)	Peak memory capped at ≤ 2.50 GB per scan
Disk Storage	1 GB free SSD storage	Outputs staged locally under `outputs/`

2 Environment Installation & Setup

Open Terminal, navigate to the project directory, and activate the virtual environment:

```
# 1. Navigate to project root directory
cd "/Users/apple/Documents/Codex/2026-09-29/untitled folder"

# 2. Activate the pre-configured virtual environment
source .venv/bin/activate

# 3. Verify pinned dependencies (if setting up on a fresh machine)
pip install -r requirements-lock.txt
```

Success Check: Running `python -V` returns `Python 3.11.x`, and `python -c "import dipy, nilearn, nibabel; print(OK)"` outputs `OK`.

3 Launching the Interactive Web Dashboard

Start the local dashboard server. It binds exclusively to `127.0.0.1` (localhost) to ensure security:

```
# Start the local dashboard server on port 8765
PYTHONPATH=. .venv/bin/python scripts/serve_dashboard.py
```

Open your browser and navigate to:

```
http://127.0.0.1:8765
```

What you will see: The dark navy **Research Workbench** containing the Overview cards, MRI Data Intake upload sections, Pipeline Control runner, and Execution History table.

4 Running Your First Preprocessing Pipeline

You can execute the pipeline through the web interface or via the command line.

Option A: Via Web Dashboard (1-Click Execution)

1. Open **http://127.0.0.1:8765**.
2. Scroll down to the **Pipeline Control** panel.
3. Select your parameters (Default: `2 subjects`, `60 volumes`, Seed `42`, `Synthetic phantom atlas`).
4. Click the blue button: **Run MRI pipeline →**.
5. The status badge turns to **RUNNING**. Wait ~4–8 seconds.
6. The table updates to **SUCCESS**. Click the run row to view Quality Control overlays and the 19,900 Fisher-z connectivity matrix.

Option B: Via Command Line (Direct Headless Runner)

Execute the production-hardened 19-stage pipeline runner directly in Terminal:

```
# Set the mandatory fail-closed cryptographic key and execute
NEUROADHD_HMAC_KEY="neuroadhd_demo_run_key" PYTHONPATH=. \
  .venv/bin/python scripts/run_poc_v2.py --output-dir outputs/demo_run
```

Successful Output: Completes all 19 stages in ~4.3 seconds with peak RSS memory ≤ 0.29 GB.

5 Where Are My Generated Outputs?

All pipeline deliverables are stored under `outputs/<run_directory>/`. Key files include:

File Name	Format	Description
`roi_CC200.csv`	CSV (200 × T)	**Primary Deliverable:** Cleaned BOLD timeseries for all 200 Craddock ROIs.
`features_19900_fisher_z.csv`	CSV (1D vector)	19,900 pairwise Fisher-z transformed Pearson correlation features.
`motion_parameters.tsv`	TSV (6 cols)	Estimated 3 translational (mm) and 3 rotational (°) head movements.
`framewise_displacement.tsv`	TSV (1 col)	Framewise Displacement (FD in mm) with 0.50 mm scrubbing indicators.
`confounds.tsv`	TSV (31 cols)	Full nuisance matrix (Friston-24 + 5 aCompCor + linear/quadratic trends).
`qc_parcellation_cc200.png`	PNG (High-Res)	Diagnostic orthographic image verifying CC200 parcels on standard brain.
`qc_tissue_masks.png`	PNG (High-Res)	Quality control overlay showing CSF, Grey Matter, and White Matter masks.
`provenance.json`	JSON (Schema)	Audit manifest with cryptographic SHA-256 hashes and tool versions.

6 Essential Troubleshooting & Verification Commands

Issue	Probable Cause	One-Line Solution

`Port 8765 in use`	Previous server process running	`lsof -ti:8765 | xargs kill -9`
`ModuleNotFoundError: src`	`PYTHONPATH` not set	Prefix command with `PYTHONPATH=.`
`HMAC key error`	Safety environment variable missing	Export key: `export NEUROADHD_HMAC_KEY="local_key"`
Run unit tests	Verify codebase integrity	`.venv/bin/pytest tests/unit -v` (175 tests pass)